

ALOK KUMAR DAS

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EDUCATION

Indian Institute of Information Technology (IIIT), Dharwad

Dharwad, Karnataka

B.Tech in Computer Science

Sept. 2023 – July 2027

- Coursework: DSA, OS, Computer Networks, DBMS, OOP, Software Engineering, DAA | **500+** problems on LeetCode, CodeChef, Codeforces

EXPERIENCE

Figital Labs

Remote

Software Development Engineer Intern – Full Stack

Jan 2026 – Apr 2026

- Architected HAQMS, a distributed platform with concurrent real-time queue tracking, RBAC, and automated slot management via Next.js, Node.js, PostgreSQL, Prisma, Socket.IO; cut scheduling time by **40%**.
- Resolved **10+ critical bugs** (N+1 queries, Socket.IO race conditions, auth vulnerabilities); authored onboarding docs reducing dev setup by **50%**.

IBM SkillsBuild – Edunet Foundation

Remote

Frontend Developer Intern

Aug. 2025 – Oct. 2025

- Built responsive React.js portfolio with optimized component architecture; achieved **95+ Lighthouse score** across desktop and mobile.

PROJECTS

Citadel – RAG-Powered Multi-Agent AI Platform | 🐙 GitHub

- Engineered a production-grade multi-agent AI platform with **8 specialized agents** (chat, coding, vision, web search, PDF **RAG**, image gen, PPT) orchestrated via LangGraph supervisor with intelligent task routing and inter-agent communication.
- Built full **RAG** pipeline using Qdrant vector search for document ingestion, embedding, and context-aware retrieval; LLM orchestration across **4 providers** (Gemini, Groq, OpenRouter, DeepSeek) behind a single unified API.
- Integrated Firebase auth cached via Redis, Razorpay billing with per-invocation credit deduction, AWS S3 presigned uploads, PDF parsing, and on-demand PPT generation.
- Deployed **5-service microservices** (Gateway, Auth, Chat, Agent, Billing) via Docker Compose with Redis 7, MongoDB 7, Qdrant; Nginx production frontend with rate-limiting and auth middleware.
- **Tech Stack:** React 19, Vite, Redux, TailwindCSS, Node.js, Express 5, LangChain, LangGraph, MongoDB, Redis, Qdrant, Firebase, Docker, AWS S3, Razorpay, Tavily, Groq, Gemini, JWT

TestGen AI – AI-Powered Automatic Test Case Generator | [Live](#) | 🐙 GitHub

- Architected a scalable AI platform auto-generating unit, integration, and acceptance test cases from source code; **60%** reduction in test-writing effort; multi-framework export (Pytest, JUnit, Jest) with self-healing support.
- Fine-tuned Qwen 1.5B via LoRA + Groq API; GitHub Actions CI/CD, CodeQL, Docker, **99%+ uptime** – Runner-Up, Agentica 2.0 Hackathon (**200+ teams**).
- **Tech Stack:** React 19, Vite, Node.js, Express, MongoDB, Docker, GitHub Actions, Groq API, JWT, CodeQL

HAQMS – Hospital Queue Management | [Live](#) | 🐙 GitHub

- Production-grade hospital platform with real-time queue via Socket.IO, RBAC, secure REST APIs, Prisma ORM migrations, JWT auth serving **3 user roles**; deployed on Vercel/Render/Neon.
- **Tech Stack:** Next.js, Node.js, Express, PostgreSQL, Prisma, Socket.IO, JWT, Vercel, Render

NativeHarvest – XAI Crop Mapping | 🐙 GitHub

- XAI framework for Indian crop mapping with PyTorch Attention-LSTM + multi-modal Sentinel-1/2 satellite fusion; **80.8% accuracy, 6.70% spatial error**; SHAP GradientExplainer interpretability across ML architectures.
- **Tech Stack:** PyTorch, Attention-LSTM, Sentinel-1/2, SHAP GradientExplainer, NumPy, Pandas, Matplotlib

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C, C++

AI/ML & LLM: LangChain, LangGraph, Groq, Gemini, OpenRouter, LLM Fine-tuning (LoRA), Qdrant, **RAG**, SHAP, PyTorch, Pandas, NumPy

Web & Frameworks: React.js, Node.js, Next.js, Express.js, Redux, TailwindCSS, REST APIs, Socket.IO, WebSockets, JWT, RBAC

Distributed Systems: Apache Kafka, Redis, Microservices, Event-Driven Architecture, System Design, Async Programming

Databases: PostgreSQL, MongoDB, Redis, Qdrant, Prisma ORM, SQL

DevOps & Cloud: Docker, Docker Compose, GitHub Actions CI/CD, AWS S3, Vercel, Nginx, Git, CodeQL, Linux

Auth & Payments: Firebase Auth, Razorpay, OAuth

Collaboration: Technical Documentation, Code Review, Agile/Scrum, Cross-functional Teamwork, Stakeholder Reporting

ACHIEVEMENTS & OPEN SOURCE

Hackathon – 1x Winner, Top 2 of 150+ Teams | TestGen AI recognized for production-grade AI system architecture

2026

SQL (Gold) – HackerRank | Earned Gold Badge for SQL mastery on HackerRank

2026

GSSoC '26 – Open Source Contributor | Merged PRs: AlphaSight AI (API crash fix, profile nav bug) and Neutralinojs CLI

2026

Sangeet 4.0 – Operations & Logistics Lead | Led coordination for **1000+** attendee cultural fest; managed end-to-end logistics

2025

Sangeet 3.0 – Event Security Lead | Coordinated security and crowd management for large-scale college cultural festival

2024